Post-Quantum Digital Signatures

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***Abstract*—The foundational security of modern Public-Key Infrastructure (PKI), reliant on algorithms like RSA and Elliptic Curve Cryptography (ECC), is critically threatened by the anticipated emergence of powerful quantum computers capable of executing Shor's algorithm. To ensure the long-term integrity and non-repudiation of digital communication, an urgent transition to Post-Quantum Cryptography (PQC) is required.**

***Index Terms*—Post Quantum Cryptography, Digital Signatures, Hash-Based Signatures, R-LWE, Lattice, Public Key Infrastructure.**

I. Introduction

DIGITAL signatures form the foundation of modern public key infrastructure, which has been deployed in many applications, from financial transactions to network security protocols like TLS. The foundational cryptographic security of established public-key algorithms, notably RSA and Elliptic Curve Cryptography (ECC), is critically compromised by the potential power of quantum computers. Their continued development has put these mathematical problems at risk, creating an urgent need to switch to Post-Quantum Cryptography systems to ensure these applications remain secure.

This review is focused on analyzing the current state of post-quantum digital signature schemes. The scope is to evaluate PQC signature schemes, their construction, the needed security evolutions, and the strategies for transitioning them into existing infrastructure through hybridization.

II. Primary Constructions, PQC

The primary constructions currently under consideration in post-quantum cryptography (PQC) focus heavily on lattice-based schemes, exemplified by TESLA#, and robust hash-based schemes, such as SPHINCS. These candidates aim to provide existential unforgeability under quantum chosen-message attacks, often incorporating features for efficiency and misuse resistance beyond classical security definitions.

A. Lattice-Based Schemes

The TESLA# digital signature scheme is founded on the Ring Learning with Errors (R-LWE) assumption, leveraging the flexibility and strong theoretical underpinnings of ideal lattices to achieve high performance and relatively small key sizes[1].

Performance and Efficiency: TESLA# improves upon its predecessors (like Ring-TESLA) by achieving significantly faster key pair generation, signing, and verification[1]. It aims for proper 128-bit post-quantum security while remaining competitive even against pre-quantum schemes.

Side-Channel Resistance: The scheme is designed for robustness against side-channel attacks by being implemented in an isochronous fashion (resistant to timing and cache attacks)[1]. A core factor enabling this performance profile is that the computationally expensive Gaussian sampling is required only for key pair generation; the signing process itself relies solely on uniform sampling, favoring high-performance isochronous signing.

Simplified Key Generation: Key generation is substantially sped up by simplifying the process: components of the private key are simple Gaussian samples and are not filtered to ensure coefficient size limitations (a requirement present in predecessors like TESLA)[1]. This change eliminates the need for maintaining complex data structures, such as priority queues, which are difficult to implement in a side-channel resistant manner.

Improved Correctness and Rejection: TESLA# addresses a flaw in the rejection conditions used by predecessor schemes that could cause valid signatures to fail verification. It resorts to the exact rejection condition requiring the hash function input to be identical during signing and verification[1]. This simplification leads to a high signing success rate, typically above 75

Security Against Weak Lattices: To mitigate the risk of an adversary exploiting a "weak" lattice instance or performing "all-for-the-price-of-one" precomputation attacks, TESLA# recommends a variant where a fresh pair of public constants is generated for every key pair from a short seed.

B. Robust Hash-Based Schemes

Hash-based signature schemes, like SPHINCS (and its successor, SPHINCS+), are sought after because their security relies primarily on the well-studied collision resistance and preimage resistance properties of cryptographic hash functions[2]. This provides a strong foundation for security against quantum adversaries.

Stateless Design: The most critical advancement in SPHINCS is its statelessness. Unlike prior practical hash-based schemes that were stateful (requiring the signer to maintain and update a secret index, failure of which could lead to security disintegration), SPHINCS avoids this requirement, making it suitable as a direct replacement for existing digital signature APIs[2].

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Efficiency Through Hyper-Trees and FTS: To achieve practicality while remaining stateless, SPHINCS introduced key ideas, such as a hyper-tree structure, and a replacement for the standard one-time signature (OTS) required at the leaves with a Few-Time Signature (FTS) scheme called HORS with trees (HORST). This combination drastically reduces the required height of the hyper-tree, which in turn greatly minimizes signature size, addressing a major practical challenge faced by earlier theoretical hash-based constructions that resulted in multi-megabyte signatures[2]. The SPHINCS-256 achieves long-term 2^{128} security and performs hundreds of signatures per second on modern CPUs. While the key sizes are small (1 KB), the signatures are large (41 KB) compared to lattice schemes.

Security Beyond Unforgeability (SPHINCS+): SPHINCS+ is explicitly designed to meet security properties that go beyond simple Existential Unforgeability (EUF-CMA), focusing on misuse resistance. SPHINCS+ achieves Message-Bound Signatures (MBS), preventing an adversary from creating a key under which two different messages verify with the same signature, assuming the underlying hash functions are collision resistant[3]. Experts also anticipate that SPHINCS+ satisfies Malicious-Strong Universal Exclusive Ownership (M-S-UEO) and Non-Re-signability (NR) due to the scheme's construction, although formal proofs under standard assumptions were pending[3].

Overall, both lattice-based schemes like TESLA# and hash-based schemes like SPHINCS represent significant advances in the PQC landscape, with TESLA# emphasizing fast, side-channel-resistant signing enabled by R-LWE, and SPHINCS prioritizing robustness and operational compatibility through its breakthrough stateless design derived from secure hash functions.

III. Signature Security

The traditional notion of Existential Unforgeability under Chosen Message Attack (EUF-CMA) is increasingly recognized as insufficient for securing complex modern cryptographic protocols, necessitating a shift toward stronger security guarantees. EUF-CMA, while foundational, only ensures that an adversary cannot construct a valid signature for a new message without the secret key, leaving protocols vulnerable to subtle, unexpected behaviors related to key management and signature manipulation.

A. EUF-CMA Insufficiency

EUF-CMA has been the standard definition for digital signatures for over 30 years. However, schemes that are provably EUF-CMA-secure can still exhibit behaviors that lead to real-world attacks in modern protocols[3,4].

Modern digital signature schemes must satisfy properties far stricter than the traditional Existential Unforgeability under Chosen Message Attack (EUF-CMA) to ensure security in complex real-world protocols[3]. Additionally, schemes may suffer from Malleability, where an adversary can modify a valid signature without affecting its verifying status, or Message/Signature Collisions, which allows a single signature to verify against multiple distinct messages, breaking the fundamental assumption that the signer knew the exact content they signed[3]. The problem of Re-signability allows an adversary to take a valid signature on an unknown message and produce a new, valid signature for that same message under their own key, compromising the protocol's integrity. These subtle failures demonstrate the critical need for PQC candidates to be scrutinized against a range of security properties that go beyond simple unforgeability[3,4].

B. Beyond Unforgeability (BUFF)

To prevent misuse, new standards focus on achieving properties beyond unforgeability (BUFF).

Exclusive Ownership (EO) is the property that a signature only verifies under a single public key[4]. Its failure is linked to Duplicate-Signature Key Selection (DSKS) attacks.

Conservative Exclusive Ownership (CEO) requires that an adversary cannot find a new public key distinct from the original that validates a signature on the original message.

Destructive Exclusive Ownership (DEO) is a stronger variant in which the resulting signature is verified against a different message[4].

Malicious-Strong Universal Exclusive Ownership (M-S-UEO), the strongest variant, requires the adversary to create a signature that satisfies two distinct public keys for two potentially distinct messages, even if the adversary chose both pairs of keys [4]. M-S-UEO is strictly stronger than S-UEO (Strong Universal Exclusive Ownership), and thus stronger than S-CEO and S-DEO.

IV. Security Models for Quantum Adversaries

Classical security definitions (such as EUF-CMA) assume that adversaries make classical queries. Quantum security requires adapting the adversary model to allow for quantum capabilities and queries[5,6].

In security proofs involving hash functions, particularly for PQC candidates like SPHINCS+, the security relies on the assumption that the hash function is ideal, often modeled as a Random Oracle (RO)[5]. In the quantum setting, this becomes the Quantum Random Oracle Model (QROM), where the adversary is allowed to issue quantum queries (superpositions of inputs) to the RO. This model is crucial for proving the security of generic constructions and compilers used to achieve quantum-secure signatures.

The BUFF transformation produces an enhanced scheme that is provably EUF-CMA secure (conditioned on the collision resistance of H) while simultaneously satisfying a suite of advanced security requirements, namely M-S-UEO, MBS, and NR[3]. Schemes like Dilithium and Picnic already implicitly incorporate elements of this binding to achieve these properties.

V. Hybrid Signature Schemes

Contemporary transition strategies for migrating to post-quantum cryptography (PQC) emphasize the use of hybrid signature schemes that combine classical and PQC primitives. This strategy mitigates unforeseen risks associated with the

relative immaturity of the PQC candidates and maintains backward compatibility with existing infrastructure and software[6,7].

Given the recent introduction of PQC primitives, the absence of extensive peer review and real-world deployment means they are currently considered higher risk, potentially harboring undiscovered vulnerabilities or subtle complexities[6]. Hybridization ensures that if either the classical algorithm (e.g., threatened by a future quantum computer) or the PQC algorithm (e.g., threatened by a classical or quantum breakthrough) remains unbroken, confidentiality or authenticity can still be ensured. The security of the hybrid scheme should ideally be reducible to the security of its component algorithms[7] to assure users that the combination builds on existing analysis and standardization efforts.

Algorithms like Dilithium are complex and supported by extensive specifications, compared with the relative simplicity of legacy algorithms like RSA[7]. If even simple classical schemes have taken time to mature in implementation practice, complex PQC schemes are logically assumed to harbor undiscovered subtleties. Hybrid schemes allow the needed transition to proceed while acknowledging the need for more time for PQC primitives to undergo pervasive cryptanalysis.

Delaying the deployment of PQC solutions risks a scenario in which the time remaining before a quantum computer is ready for cryptanalysis is less than the time needed for the transition, famously summarized by Mosca's equation[7]. For large systems where acquisition and integration times are measured in years, the need for quantum resistance now often outranks the desire for more algorithm observation time, pushing organizations toward immediate, hybridized transitions.

The practical success of these strategies ultimately depends on how far developers can push existing cryptographic APIs to accommodate the larger keys and signatures of PQC schemes while minimizing the performance impact and ensuring graceful degradation (ignoring unrecognized data) rather than failure.

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